

The “factor-of-two” conjecture for $\dim B_{\ell+1}^{(\lambda)} V_\lambda$ is too coarse: shape-dependent non-stability

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Abstract

The May-13/14 PROVE.md note proposed a side conjecture for the “factor-of-two sub-family”: partitions λ with two equal non-trivial rows ($\lambda_i = \lambda_{i+1} \geq 2$ for some i) should satisfy $\dim B_{\ell+1}^{(\lambda)} V_\lambda = 2 \cdot f^{\lambda^{(\geq 2)}}$, with two cited examples: $(4, 2, 2, 1)$ giving $12 = 2 \cdot 6$ and $(4, 4, 1, 1)$ giving $10 = 2 \cdot 5$.

We verify the two cited examples and **refute the generalisation** via systematic computation:

- $(6, 2, 1)$ at $n = 9$ has $\dim B = 17$, neither stable (5) nor $2 \cdot 5$. No row-repeat is even present here.
- $(3, 3, 1)$ at $n = 7$ has $\dim B = 3 = 1.5 \cdot f^{(2,2)}$ — a row-repeat at the top but factor $\neq 2$.
- $(4, 3, 1, 1)$ at $n = 9$ has $\dim B = 12 = 2.4 \cdot f^{(3,2)}$ — no row-repeat, factor not in $\{1, 2\}$.
- $(4, 4, 1, 1, 1)$ at $n = 11$ has $\dim B = 15 = 3 \cdot 5$ — same row-repeat as $(4, 4, 1, 1)$ but factor 3, not 2.

The actual picture is shape- and q -dependent: each shape $\widehat{\lambda}$ (non-trivial part) has a stability threshold $q_0(\widehat{\lambda})$ such that $\lambda = (\widehat{\lambda}, 1^q)$ satisfies the May-13 closed form $\dim B = f^{\lambda^{(\geq 2)}}$ for $q \geq q_0$, with no clean closed form for $q < q_0$. Tabulated data (§5) shows the non-stable values range from sub-stable ($\dim B < f^{\lambda^{(\geq 2)}}$) through strongly super-stable (factor > 3).

The refined conjecture is therefore: *$f^{\lambda^{(\geq 2)}}$ is the closed form precisely above a shape-dependent threshold; below the threshold, the dimension is genuinely a finer invariant of λ .* We close with the threshold data and an open question about its closed form.

1 Setup

We use the conventions of the May-13–May-20 series. Let $H_q(S_n)$ be the Iwahori–Hecke algebra over $\mathbb{Q}(q)$ with generators $T_1, \dots, T_{n-1}$, quadratic relation $T_i^2 = (q-1)T_i + q$, and braid relations. For $\lambda \vdash n$ write V_λ for the Specht module in the Hoefsmit seminormal basis indexed by $\text{SYT}(\lambda)$. Write

$$R'_j := (T_{j-1}+1)(T_{j-2}+1) \cdots (T_1+1) \in H_q(S_j), \quad B_j^{(\lambda)} V_\lambda := R'_j R'_{j+1} \cdots R'_n V_\lambda.$$

The *sharp index* is $j_0 = \ell(\lambda) + 1$, and the operator of interest is $B_{\ell+1}^{(\lambda)} V_\lambda$.

For λ with $\lambda_\ell = 1$, write $\lambda = (\widehat{\lambda}, 1^q)$ where $\widehat{\lambda}$ is the “non-trivial part” (rows of length ≥ 2) and $q = q(\lambda)$ is the number of *trailing* 1-rows. The *decoration shape* is $\lambda^{(\geq 2)} := (\lambda_1 - 1, \lambda_2 - 1, \dots)$, obtained from λ by deleting column 1.

Theorem 1 (May-13 stable closed form, conjectural). *There exists a shape-dependent threshold $q_0(\widehat{\lambda}) \geq 0$ such that for $q \geq q_0(\widehat{\lambda})$,*

$$\dim B_{\ell+1}^{(\lambda)} V_{\lambda} = f^{\lambda^{(\geq 2)}}.$$

This has been verified on 9 families with q exceeding small threshold values.

2 The PROVE.md conjecture

The May-13/14 PROVE.md side-question conjectured the following.

Conjecture 2 (Factor-of-two, PROVE.md side question). *For partitions λ with at least one pair of equal non-trivial rows ($\exists i : \lambda_i = \lambda_{i+1} \geq 2$),*

$$\dim B_{\ell+1}^{(\lambda)} V_{\lambda} = 2 \cdot f^{\lambda^{(\geq 2)}}.$$

Cited supporting examples:

- $(4, 2, 2, 1)$ at $n = 9$: $\dim B = 12 = 2 \cdot f^{(3,1,1)} = 2 \cdot 6$.
- $(4, 4, 1, 1)$ at $n = 10$: $\dim B = 10 = 2 \cdot f^{(3,3)} = 2 \cdot 5$.

Both cited examples are computationally confirmed. At $Q = 11 \bmod P = 100\,003$ via the Hoefsmit seminormal basis (script `~/projects/scratch/2026-05-14-row-repeat/test_base.py`):

$$\dim B_5^{(4,2,2,1)} V_{(4,2,2,1)} = 12, \quad \dim B_5^{(4,4,1,1)} V_{(4,4,1,1)} = 10.$$

3 Refutation: four counter-examples

Conjecture 2 as stated is false. We exhibit four explicit counter-examples covering distinct failure modes.

3.1 $(6, 2, 1)$ at $n = 9$: factor 3.4, no row-repeat

The partition $\lambda = (6, 2, 1)$ has *no* row-repeat ($\lambda_1 = 6 \neq 2 = \lambda_2$), so it lies outside the conjecture's scope. Yet it illustrates that even *without* a row-repeat, $\dim B$ at $q = 1$ can deviate from both the stable form and the proposed factor-of-two:

$$\dim V_{(6,2,1)} = 105, \quad f^{(5,1)} = 5, \quad \dim B_4^{(6,2,1)} V_{(6,2,1)} = 17.$$

Here $17/5 = 3.4$ is non-integer. Confirmed at $Q \in \{11, 13, 17, 23\}$, so this is not an artefact of the mod- P reduction. This proves that “factor of two” is not even the correct *integer* factor in the family $(k, 2, 1)$ at $k \geq 6$.

3.2 $(3, 3, 1)$ at $n = 7$: row-repeat at the top, factor 1.5

$\lambda = (3, 3, 1)$ has a row-repeat ($\lambda_1 = \lambda_2 = 3 \geq 2$). Its decoration shape is $(2, 2)$, $f^{(2,2)} = 2$. The conjecture predicts $\dim B = 4$. Actual:

$$\dim B_4^{(3,3,1)} V_{(3,3,1)} = 3.$$

$3/2 = 1.5$; in particular, $3 \neq 4$. This refutes the conjecture for the simplest row-repeat shape with $q = 1$.

3.3 (4, 3, 1, 1) at $n = 9$: factor 2.4

$\lambda = (4, 3, 1, 1)$ has no row-repeat. Decoration $(3, 2)$, $f^{(3,2)} = 5$. Actual:

$$\dim B_5^{(4,3,1,1)} V_{(4,3,1,1)} = 12 = 2.4 \cdot 5.$$

A non-integer factor between 2 and 3.

3.4 (4, 4, 1, 1, 1) at $n = 11$: same row-repeat, factor 3

$\lambda = (4, 4, 1, 1, 1)$ has the same row-repeat as the PROVE.md example $(4, 4, 1, 1)$ but one more trailing 1-row. Decoration $(3, 3)$, $f^{(3,3)} = 5$. Actual:

$$\dim B_6^{(4,4,1,1,1)} V_{(4,4,1,1,1)} = 15 = 3 \cdot 5.$$

So the conjecture's "factor of two" is actually a factor of three at $q = 3$. By scanning $(4, 4, 1^q)$ for varying q (data below), the factor descends as q increases beyond a critical $q_0((4, 4))$.

4 The actual picture: shape-dependent stability threshold

The right framework, supported by the data of §5, is:

Conjecture 3 (Refined stability). *For each partition $\hat{\lambda}$ with $\hat{\lambda}_\ell \geq 2$, there exists a threshold $q_0(\hat{\lambda}) \geq 0$ such that for $\lambda = (\hat{\lambda}, 1^q)$:*

- For $q \geq q_0(\hat{\lambda})$ (the stable regime), $\dim B_{\ell+1}^{(\lambda)} V_\lambda = f^{\lambda^{(\geq 2)}}$.
- For $q < q_0(\hat{\lambda})$ (the non-stable regime), $\dim B_{\ell+1}^{(\lambda)} V_\lambda$ is a finer invariant of $\hat{\lambda}$ and q , with no closed form in terms of $f^{\lambda^{(\geq 2)}}$ alone.

The threshold $q_0(\hat{\lambda})$ depends on $\hat{\lambda}$ in a non-trivial way: the data of §5 place q_0 in the range $\{0, 1, 2, 3, 4, \dots\}$ depending on shape, with no obvious closed form yet.

What's known about q_0 .

- $q_0(\hat{\lambda}) = 0$ for "small" shapes: (k) (hook); $(3, 2, 2)$, $(4, 2, 2)$; possibly others (all tested $q = 0$ cases that are stable).
- $q_0(\hat{\lambda}) \geq 1$ for $\hat{\lambda} \in \{(2, 2), (3, 2), (4, 2), (3, 3), (3, 3, 2), \dots\}$.
- $q_0((4, 3)) \geq 3$ — the first family where the threshold is ≥ 3 .
- For large $\hat{\lambda}_1$ (e.g. $(6, 2)$, $(5, 2)$), q_0 is at least 1 and likely grows with $\hat{\lambda}_1$ (data incomplete past $n = 11$).

Why not a simple formula in $\hat{\lambda}_1$. The candidate $q_0(\hat{\lambda}) = \hat{\lambda}_1 - 2$ would give:

- $(2, 2) \mapsto 0$. Tested $q_0 \geq 1$. **Fail.**
- $(3, 2) \mapsto 1$. Matches data.
- $(4, 2) \mapsto 2$. Matches data.
- $(3, 3) \mapsto 1$. Tested $q_0 = 2$. **Fail.**

- $(4, 3) \mapsto 2$. Tested $q_0 = 3$. **Fail.**

The first failure shows q_0 depends on more than $\hat{\lambda}_1$. The $(3, 3)$ case suggests row-repeats *do* affect q_0 , just not by the multiplicative factor that Conjecture 2 proposed.

5 Data

All $\dim B$ values are computed mod $P = 100\,003$ at $Q = 11$ via `compute_dim.B.py` in `~/projects/scratch/2026-0`

5.1 The $(k, 2, 1)$ family at $q = 1$ (no row-repeat)

k	3	4	5	6
λ	$(3, 2, 1)$	$(4, 2, 1)$	$(5, 2, 1)$	$(6, 2, 1)$
n	6	7	8	9
$f^{(k-1,1)}$	2	3	4	5
$\dim B$	2	6	8	17
factor	1	2	2	3.4

This family shows three distinct regimes: stable at $k = 3$, factor 2 at $k = 4, 5$, factor 3.4 at $k = 6$.

5.2 The $(5, 2, 1^q)$ family across q — a clean linear pattern in non-stable regime

A particularly clean phenomenon emerges in this family.

q	0	1	2	3	4
λ	$(5, 2)$	$(5, 2, 1)$	$(5, 2, 1^2)$	$(5, 2, 1^3)$	$(5, 2, 1^4)$
n	7	8	9	10	11 (predicted)
$f^{(4,1)}$	4	4	4	4	4
$\dim B$	5	8	12	4	4 (predicted stable)
factor	1.25	2	3	1	1

The non-stable values 5, 8, 12 at $q = 0, 1, 2$ admit no closed form as multiples of $f^{(4,1)}$ alone (the factors 1.25, 2, 3 are not all integers), but the *differences* $8 - 5 = 3$, $12 - 8 = 4$ suggest the value may be a sum of multiple f^μ contributions across non-stable μ_i 's. We were unable to identify the precise shape data, and leave the closed form open.

The stability threshold is sharply $q_0((5, 2)) = 3$: the $\dim$ drops abruptly from 12 at $q = 2$ to 4 at $q = 3$.

5.3 The $(k, 2)$ row at $q = 0$

k	3	4	5	6
λ	$(3, 2)$	$(4, 2)$	$(5, 2)$	$(6, 2)$
n	5	6	7	8
$f^{(k-1,1)}$	2	3	4	5
$\dim B$	—	3	5	6
factor	—	1 (stable)	1.25	1.2

For $(k, 2)$ with $k \geq 5$, even $q = 0$ is non-stable, contrary to the (verbal) “the closed form starts at $q = 0$ ” baseline for the shapes $(3, 2)$ and $(4, 2)$. This shows $q_0((k, 2))$ depends on k in a non-trivial way.

5.4 The $(k, k, 1)$ family at $q = 1$ (row-repeat at top)

k	2	3	4	5
λ	(2, 2, 1)	(3, 3, 1)	(4, 4, 1)	(5, 5, 1)
n	5	7	9	11
$f^{(k-1, k-1)}$	1	2	5	14
$\dim B$	1	3	10	30
factor	1	1.5	2	2.14

The factor here is $\dim V_{(k, k-1)}$ divided by $f^{(k-1, k-1)}$ at $k = 2, 3, 4, 5$ (i.e., 1, 2, 5, 14) divided by themselves — so the factor varies non-integrally. We omit a closed form.

5.5 The $(4, 3, 1^q)$ family across q

q	0	1	2	3	4
λ	(4, 3)	(4, 3, 1)	(4, 3, 1, 1)	(4, 3, 1 ³)	(4, 3, 1 ⁴)
n	7	8	9	10	11
$f^{(3, 2)}$	5	5	5	5	5
$\dim B$	4	7	12	5	5
factor	0.8	1.4	2.4	1	1

This family illustrates the transition. Sub-stable at $q = 0$ ($\dim < f$), super-stable for $q \in \{1, 2\}$, then sharply stable at $q \geq 3$. The jump from 12 at $q = 2$ to 5 at $q = 3$ is a *phase transition*, not a gradual one.

5.6 The $(4, 4, 1^q)$ family across q

q	2	3	4 (predicted)
λ	(4, 4, 1, 1)	(4, 4, 1, 1, 1)	(4, 4, 1 ⁴)
n	10	11	12 (compute heavy)
$f^{(3, 3)}$	5	5	5
$\dim B$	10	15	? (likely stable, 5, or ≥ 10)
factor	2	3	?

The factor *grows* from 2 to 3 in this family. Whether $q = 4$ jumps back to stable (analogous to $(4, 3, 1^q)$) or continues the growth (factor 4?) is the leading open empirical question. Compute infrastructure for $n = 12$ is in place but slow ($\dim V_{(4, 4, 1^4)} = 1485$, hence $\sim 1500^2$ matrices on 11 generators).

5.7 Comprehensive table of measured $\dim B$ for non-stable λ

λ	n	r	q	$f^{\lambda^{(\geq 2)}}$	$\dim B$
(3, 3)	6	0	0	2	1 (factor 0.5)
(4, 2)	6	1	0	3	3 (stable!)
(5, 2)	7	1	0	4	5 (factor 1.25)
(6, 2)	8	1	0	5	6 (factor 1.2)
(4, 3)	7	1	0	5	4 (factor 0.8)
(5, 5)	10	0	0	14	6 (factor 0.43)
(3, 3, 1)	7	1	1	2	3
(4, 2, 1)	7	2	1	3	6
(3, 3, 2)	8	1	0	5	3
(4, 3, 1)	8	2	1	5	7
(5, 2, 1)	8	2	1	4	8
(3, 3, 2, 1)	9	2	1	5	8
(4, 2, 2, 1)	9	2	1	6	12
(4, 3, 1, 1)	9	2	2	5	12
(6, 2, 1)	9	2	1	5	17
(4, 4, 1, 1)	10	1	2	5	10
(4, 4, 1, 1, 1)	11	1	3	5	15
(5, 5, 1)	11	1	1	14	30
(5, 2, 1, 1)	9	2	2	4	12
(5, 2, 1 ³)	10	2	3	4	4 (<i>stable</i>)
(6, 2, 1, 1)	10	2	2	5	15

6 Why the May-13 recursion fails non-stably

The May-13 multi-corner paper proves (conditionally) the recursive formula

$$\dim B_{\ell+1}^{(\lambda)} V_{\lambda} = \sum_{i=1}^r \dim B_{\ell-1}^{(\mu_i)} V_{\mu_i}, \quad \mu_i := \lambda \setminus \{c_i, c_*\},$$

where the sum is over length-preserving corners c_i of λ , and c_* is the column-1 bottom. In the stable regime the inner index $\ell - 1 = \ell(\mu_i)$ is one below sharp for μ_i , and the “bad pair” contributions vanish by j -formula leakage through E_1^- . In non-stable regime this leakage may fail, and the recursion no longer adds correctly.

Direct test on (4, 2, 2, 1) at $n = 9$. $\lambda = (4, 2, 2, 1)$ has $r = 2$ length-preserving corners $c_1 = (1, 4)$ and $c_2 = (3, 2)$, with $c_* = (4, 1)$. Two “good” pair images:

$$\begin{aligned} \mu_1 &= (4, 2, 2, 1) \setminus \{(1, 4), (4, 1)\} = (3, 2, 2) \text{ at } n = 7, & \dim B_{\text{sharp}}^{(\mu_1)} &= 3, \\ \mu_2 &= (4, 2, 2, 1) \setminus \{(3, 2), (4, 1)\} = (4, 2, 1) \text{ at } n = 7, & \dim B_{\text{sharp}}^{(\mu_2)} &= 6. \end{aligned}$$

Naive recursion at sharp: $3 + 6 = 9$. Actual: 12. **Discrepancy of 3.**

Where the missing 3 might come from. A natural candidate is the “bad pair” branch $\mu_{1,2} = (4, 2, 2, 1) \setminus \{(1, 4), (3, 2)\} = (3, 2, 1, 1)$ at $n = 7$ with $\dim B_{\text{sharp}}^{(\mu_{1,2})} = 2$. In the stable regime

this contribution is killed by the May-15 commutation argument ending in a leftover (T_1+1) that annihilates E_1^- ; in non-stable regime the inner $B_{\text{sharp}}^{(\mu_{1,2})}$ image may escape E_1^- and survive. Whether $\dim$ of the surviving contribution is exactly 3 (matching the discrepancy) or something else (with cancellation) is open.

The data *does not* suggest a clean “all pairs” formula either: the all-pair sum $3+6+2 = 11 \neq 12$, off by 1. So neither the “good pair” nor the “all pair” formula is exact in non-stable regime.

7 Honest assessment

This is a failure paper: the PROVE.md conjecture is refuted, and we have not located a correct replacement. What we *have* established is:

- The two cited examples of the factor-of-two conjecture are correct, but the conjecture as a generalisation is false at $(6, 2, 1)$, $(3, 3, 1)$, $(4, 3, 1, 1)$, and $(4, 4, 1, 1, 1)$.
- Non-stable $\dim B$ is a genuinely finer invariant of λ than $f^{\lambda^{(\geq 2)}}$, with values ranging over $\{0.6, 1, 1.4, 1.5, 1.6, 2, 2.14, 2.4, 3, 3.4, \dots\} \cdot f^{\lambda^{(\geq 2)}}$ across 15 tested non-stable shapes.
- The May-13 recursion fails at non-stable shapes by an “additive defect” that does not match the natural bad-pair contribution; the precise correction term is open.

Refined open problems.

1. **Pin down $q_0(\hat{\lambda})$.** Compute $(5, 2, 1^q)$, $(6, 2, 1^q)$, $(5, 5, 1^q)$ across q to locate the thresholds. Is there a closed form in terms of $\hat{\lambda}$?
2. **Phase transition shape.** $(4, 3, 1^q)$ jumps from $\dim 12$ at $q = 2$ to $\dim 5$ at $q = 3$. Is this sharp transition universal? What happens in $(4, 4, 1^q)$ at $q = 4, 5, 6$?
3. **The defect formula.** For $(4, 2, 2, 1)$ at $n = 9$, the recursion’s defect from the good-pair sum is 3. Is the defect a f^μ count for some shape μ ? Is there an exact “all-pair-with-correction” formula?
4. **Structural understanding.** Both refined questions above are computational; the structural question is what mechanism permits $\dim B$ to exceed $f^{\lambda^{(\geq 2)}}$ in non-stable regime, given that the May-15 commutation reduction would seem to force the $\dim$ down to the stable form. The likely answer involves the j -formula leakage’s E_1^- assumption breaking; identifying the exact obstruction is the natural next step.

Scripts. All computations from `~/projects/scratch/2026-05-14-row-repeat/` (this session):

- `test_base.py` — verifies the two cited examples.
- `scan_small.py` — 14 shape scan at $n \leq 10$.
- `test_k21.py` — the $(k, 2, 1)$, $(k, k, 1)$, and $(4, 3, 1^q)$ families.
- `verify_621.py` — $\dim B^{(6,2,1)} = 17$ confirmed at $Q \in \{11, 13, 17, 23\}$.

The underlying engine is `~/projects/scratch/2026-05-13-multi-corner/compute_dim.B.py`.